PRACTICE PROGRAMS – 04.08.2026
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SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

- Q1 Matrix multiplication 10 marks
- Q2 . Largest number 10 marks
- Q3 Smallest number 10 marks
- Q4 . Nth largest number 10 marks
- Q5 Palindrome number/string 10 marks
- Q6 Removing duplicates in array 10 marks

6/6 answered

Q1 **Matrix multiplication** 10 marks

Matrix multiplication

Your Code

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        if (!sc.hasNextInt()) return;

        int r1 = sc.nextInt();
        int c1 = sc.nextInt();
        int[][] a = new int[r1][c1];
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c1; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        int r2 = sc.nextInt();
        int c2 = sc.nextInt();
        int[][] b = new int[r2][c2];
        for (int i = 0; i < r2; i++) {
            for (int j = 0; j < c2; j++) {
                b[i][j] = sc.nextInt();
            }
        }
        if (c1 != r2) {
            System.out.println("Multiplication not possible.");
            return;
        }

        int[][] c = new int[r1][c2];
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                for (int k = 0; k < c1; k++) {
                    c[i][j] += a[i][k] * b[k][j];
                }
            }
        }

        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                System.out.print(c[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Input

```
2 3
1 2 3
4 5 6
3 2
```

Output

```
31 19
85 55
```

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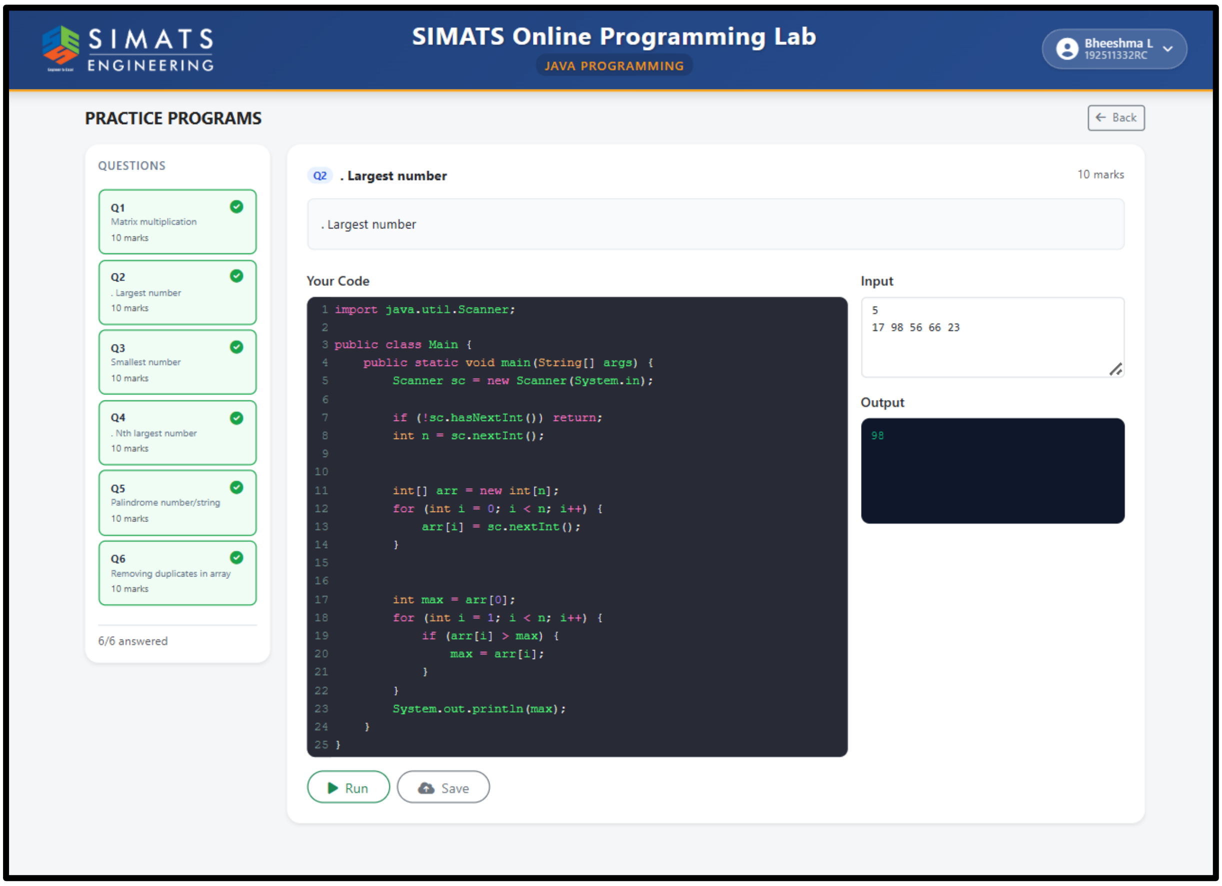
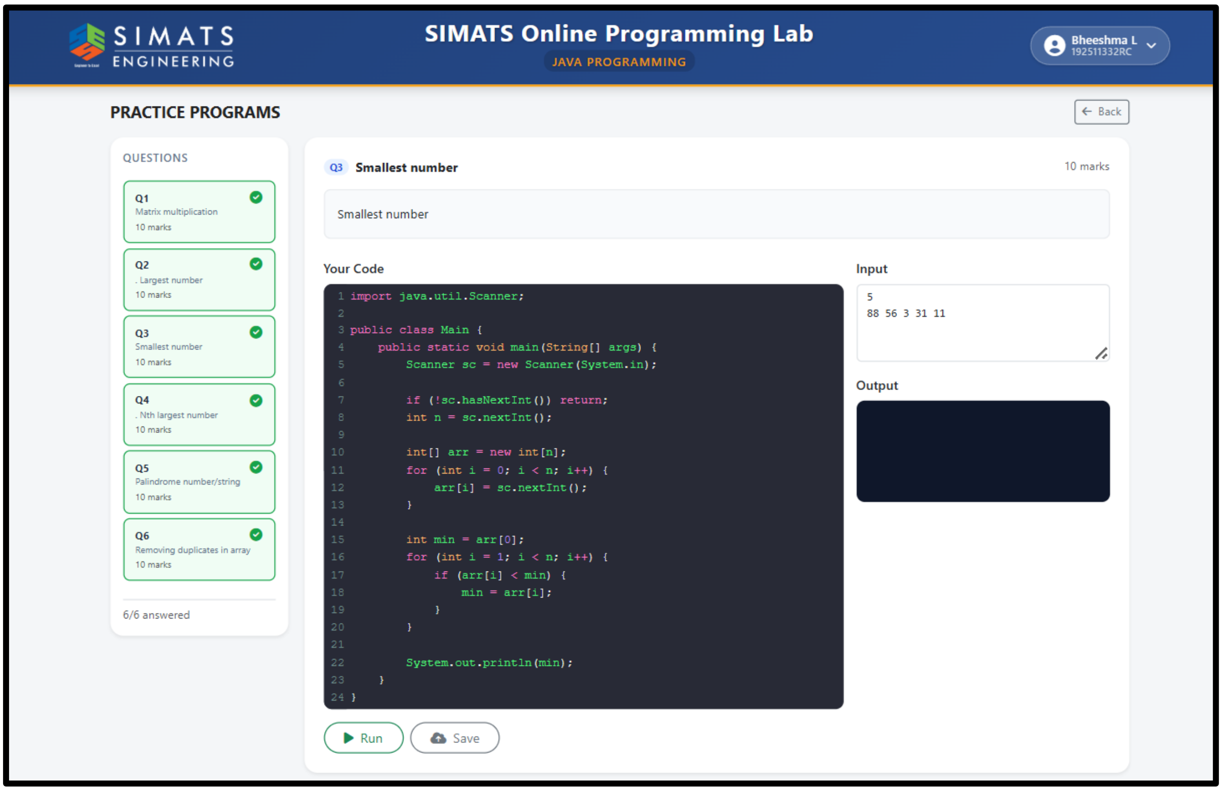



```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        if (!sc.hasNextInt()) return;
        int n = sc.nextInt();


        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }



        int max = arr[0];
        for (int i = 1; i < n; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }
        System.out.println(max);
    }
}
```




```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        if (!sc.hasNextInt()) return;
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int min = arr[0];
        for (int i = 1; i < n; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
        }

        System.out.println(min);
    }
}
```

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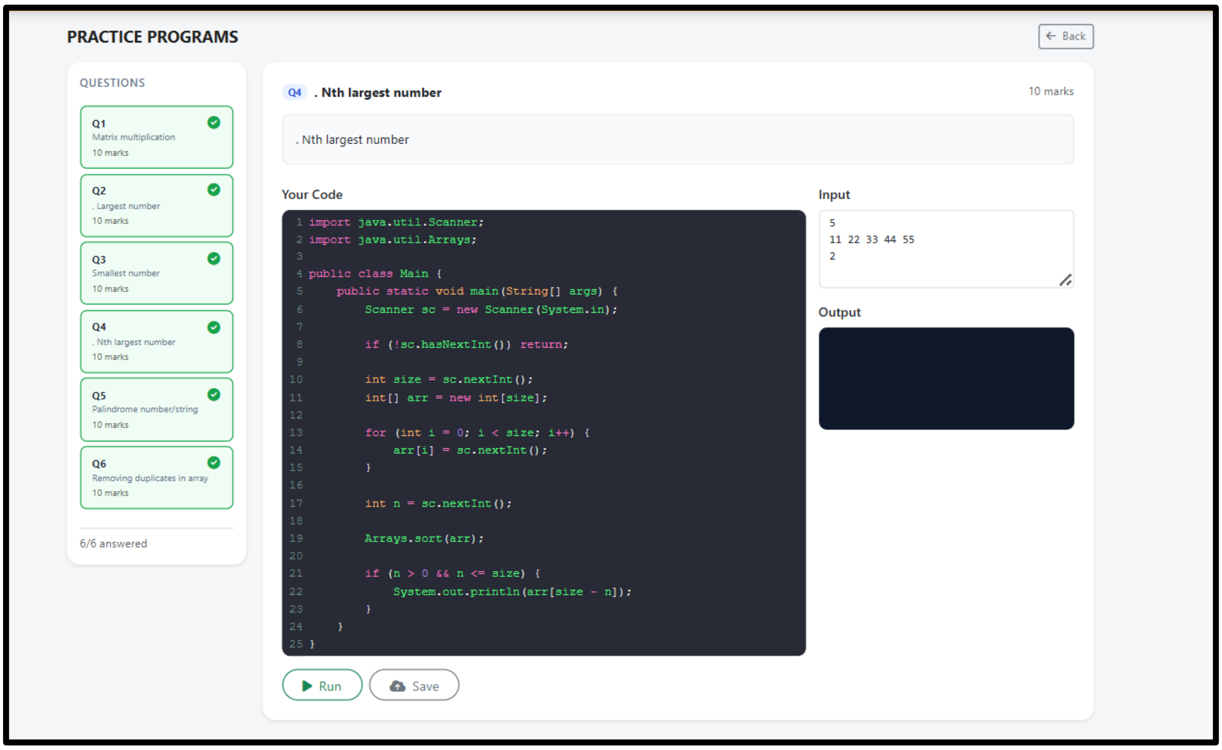

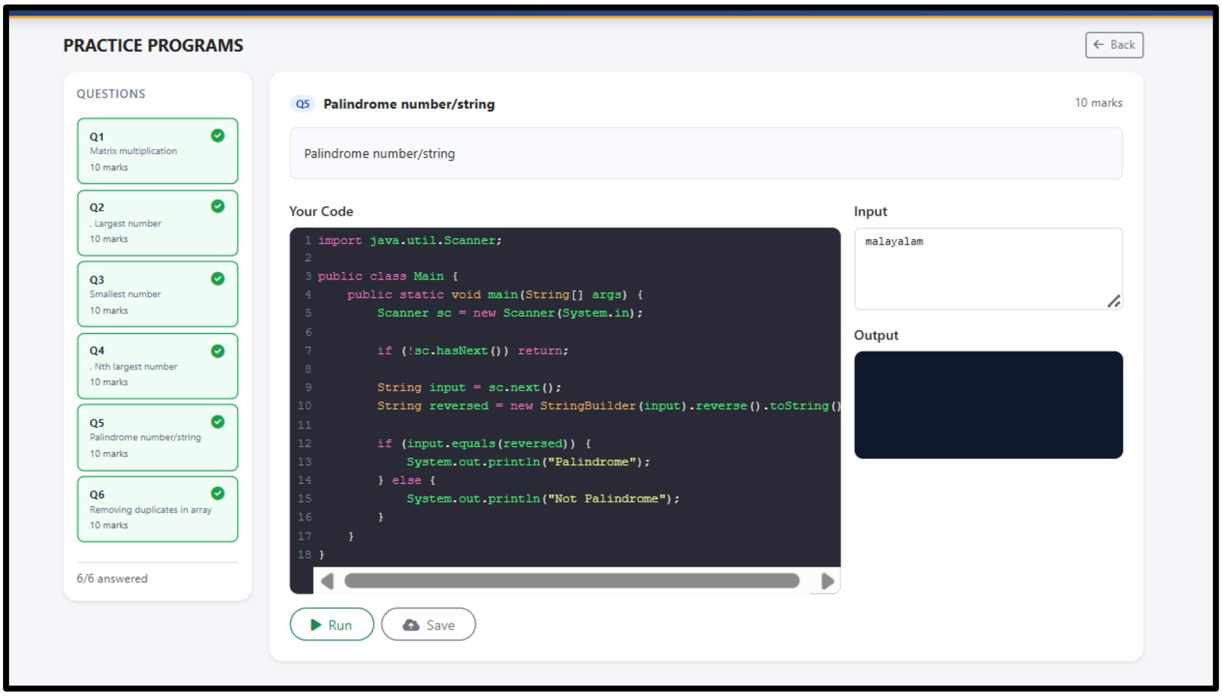

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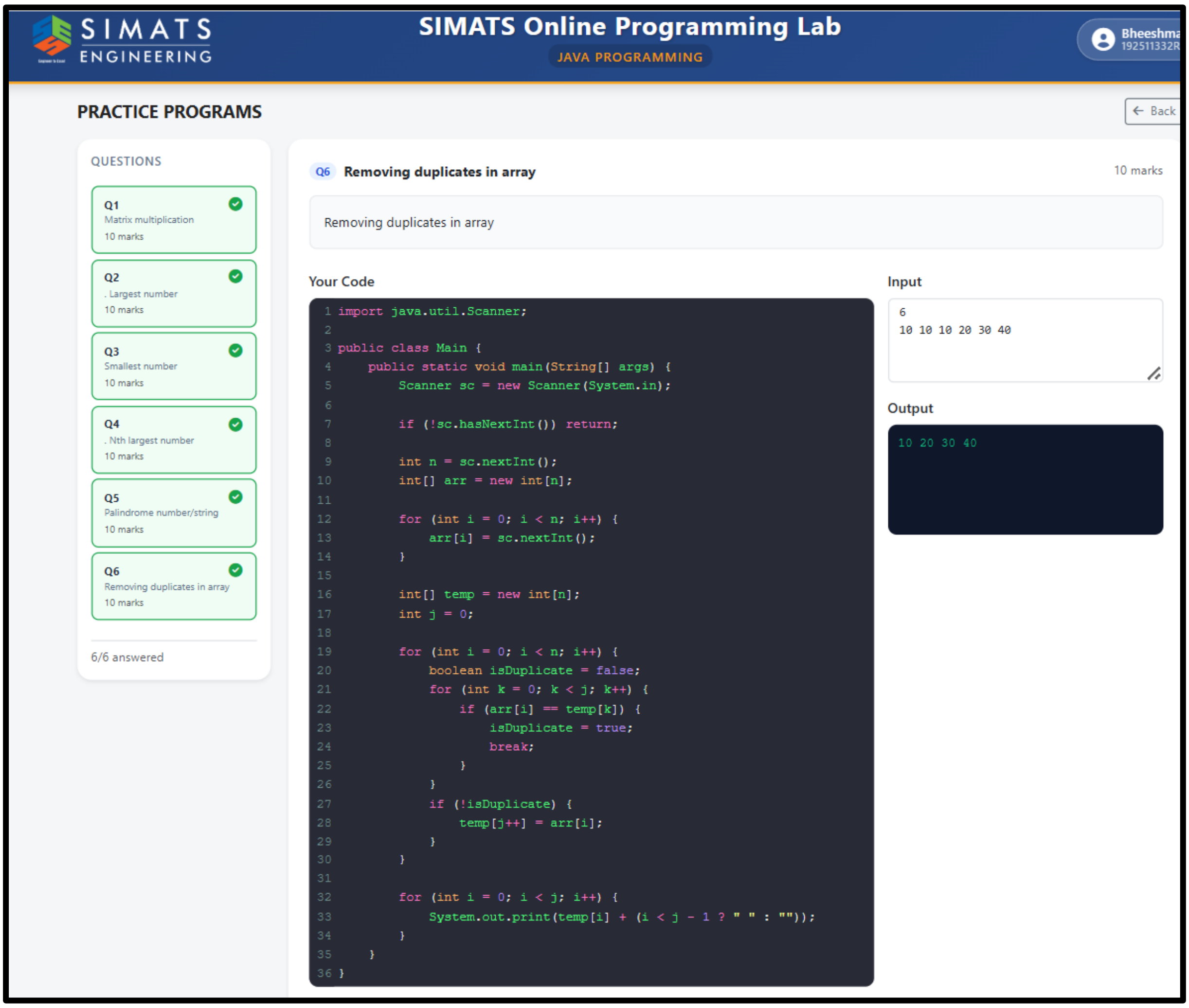


```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        if (!sc.hasNextInt()) return;

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int[] temp = new int[n];
        int j = 0;

        for (int i = 0; i < n; i++) {
            boolean isDuplicate = false;
            for (int k = 0; k < j; k++) {
                if (arr[i] == temp[k]) {
                    isDuplicate = true;
                    break;
                }
            }
            if (!isDuplicate) {
                temp[j++] = arr[i];
            }
        }

        for (int i = 0; i < j; i++) {
            System.out.print(temp[i] + (i < j - 1 ? " " : ""));
        }
    }
}
```